Richard Mitic

Senior engineer and researcher with 14 years experience specializing in audio, signal processing, embedded programming, and streaming media. Currently working towards a PhD researching how music producers use stereo field.

Email: richard.h.mitic@gmail.com

Phone: +46 073-086 44 65

Site: https://mitic.se

ORCID: 0000-0003-3232-191X

Experience

Mälardalens Universitet, SE, 2022-present

Doctoral Candidate, licentiate expected October 2026

Overarching research question:

How do music producers use stereo field?

Publications

- R. Mitic and A. Rossholm, ‘A Systematic Literature Review of Methods to Describe the Stereo Image of Recorded Music’, Journal on Audio, Speech, and Music Processing, vol. 2026, no. 1, p. 29, Jun. 2026, doi: 10.1186/s13636-026-00463-4.
- R. Mitic and A. Rossholm, ‘Towards an objective comparison of panning feature algorithms for unsupervised learning’, in Proceedings of the 28th International Conference on Digital Audio Effects (DAFx25), Ancona, IT, Sept. 2025. [Online]. Available: https://dafx.de/paper-archive/2025/DAFx25_paper_26.pdf
- A. Franklin, R. Mitic, D. Hedin, R. Lindell, and H. Frisk, ‘Development of a Web-Based Real-Time Distributed Reverberation Chamber for Live Concert Applications’, Journal of The Audio Engineering Society, vol. 73, no. 12, pp. 872–881, Mar. 2025, doi: 10.17743/jaes.2022.0245.

Spotify, Stockholm, March 2016 - April 2024

A multitude of projects centred around audio/video files, embedded programming, and research

Languages and tools:

- C, Rust, Python, C++, Java, Scala, Javascript
- Git, CMake, FFmpeg, Pyav, Docker, BigQuery, Dataflow, Xcode, Android NDK, Alsa, AndroidAudio, QT, CoreAudio

Media Tech Researcher/Engineer, June 2022 - April 2024

PhD research project with Mälardalens Universitet investigating the stereo images of recorded music

- Invention of signal processing techniques to quantify the stereo image of music
- Analysis of large cohorts of tracks via machine learning to find interesting features, trends, clusters, and general oddities
- Use of "big data" frameworks to perform DSP efficiently at scale
- Advising the company on how to maximize audio quality

C/C++/Rust Engineer, March 2021 - June 2022

Audio/video transcoding and media analysis tools

- Audio and video transcoding systems that run efficiently at the scale of Spotify
- Analysis tools for compressed media files and signal processing of their uncompressed counterparts

- Customization of open-source tools (ffmpeg, pyav)
- Proprietary audio and video encoding tools
- Analysis of weird and wonderful broken media files and parsers

Senior Engineer, April 2017 - March 2021

Spotify-branded hardware products (e.g. Car Thing)

- Embedded voice control SDK
- Real-time safe audio playback, mixing, and recording for embedded Android
- Noise cancellation in microphone signals
- Android, Linux, and QT applications for audio playback, voice control, user interface, and over-the-air updates

Developer, March 2016 - April 2017

Spotify Connect embedded playback library (eSDK)

- Customization and development of eSDK for various hardware partners (most notably Sonos)
- Automated testing of hardware products

Ericsson Research, Stockholm, 2013 - 2016

Experienced Researcher in Visual Technology, primarily focusing on MPEG DASH and WebRTC

- Developed audio/video streaming algorithms that were approved for patents
- Built test suites for MPEG-DASH, encompassing content generation and consumption
- Developed several media-file-related frameworks and applications internal use and external demonstration
- Built a head-mounted display video renderer module for OpenWebRTC
- Analysed and categorized delays in DASH media chains
- Contributed to the standardization of DASH and the MPEG File Format

Languages and tools used:

- Python, C, Javascript
- Visual Studio, Gstreamer, OpenGL

Dolby Laboratories, Nürnberg, 2012

Intern in audio compression software engineering

- Created bitstream analysis tools for a proprietary audio codec
- Contributed to the corresponding bitstream encoder
- Implemented bug fixes for in-house development tools

Languages and tools used:

- C, Python
- Visual Studio, Perforce

Education

University of Glasgow, UK, 2008-2013

Master of Engineering in “Electronics with Music” with First Class honours

- Selected modules
  - Audio Programming
  - Audio and Video Processing and Encoding
  - Data Acquisition for Musical Processing
  - Real-time Embedded Programming
  - Acoustics
  - Mathematics
  - Sonic Arts
- Selected projects

    - Automated image acquisition of nanopattern cell cultures using an optical microscope
    - Music visualisation software
    - Live music performances using Max/MSP
    - Non-contact temperature sensors for racecar tyres
- Honours and awards
    - IET Prize for outstanding academic achievement, 2013
    - Highest GPA in music department, 2011
    - Excellence in Electronics bursary, 2010/11
    - Engineering Excellence list, all semesters
- Extracurricular
    - Lead tenor saxophonist in Big Band
    - Founding member of Contemporary Music Ensemble
    - Webmaster for Filmmaking Society.

Other

Languages spoken: Native English, Basic Swedish, Basic German

Music: https://soundcloud.com/richardmitic

Code: https://github.com/richardmitic

References available on request